Jülide Kahkeci

maskoun@ualberta.ca | Edmonton, Alberta | ORCID | Scholar | Scopus | Web of Science | LinkedIn

EDUCATION

Sep 2021–Sep 2025 **PhD in Environmental Engineering**
University of Alberta · Edmonton, AB · GPA 4.00/4.00
Thesis: Biochar-Supported Bismuth Tungstate Photocatalysts for the Removal of Emerging Contaminants from Wastewater.

Sep 2018–Aug 2020 **MSc in Materials Science and Nanotechnology**
Bilkent University · Ankara, Turkey · GPA 3.79/4.00
Thesis: Microfluidic waveguides and solution lasers of colloidal semiconductor quantum wells.

Sep 2014–Jun 2018 **BSc in Chemical Engineering**
Middle East Technical University · Ankara, Turkey · GPA 3.72/4.00 · Graduated first in rank
Research: Organic solvent nanofiltration membrane synthesis and application in wastewater treatment.

RESEARCH EXPERIENCE

Nov 2025–Present **Postdoctoral Research Fellow, Water and Wastewater Treatment**
University of Alberta · Edmonton, AB
• Research forestry-waste biochar for oil sands process water in pilot-scale columns and constructed wetlands; develop solar/UV-driven photocatalytic reactors and biochar/hydrochar adsorbents. Co-supervise three PhD students and collaborate with industry partners.

Sep 2021–Sep 2025 **Graduate Research Assistant, Water and Wastewater Treatment**
University of Alberta · Edmonton, AB
• Designed and characterized biochar-supported photocatalysts, assessed toxicity and leaching in real water matrices, developed HPLC methods, and achieved a 74-fold improvement in photocatalytic activity.

Sep 2020–Aug 2021 **Graduate Research Assistant, Green Hydrogen Production**
King Abdullah University of Science and Technology · Jeddah, Saudi Arabia
• Evaluated solar- and wind-powered PEM and alkaline electrolysis using mass and energy balances and techno-economic assessment.

Sep 2018–Aug 2020 **Graduate Research Assistant, Microfluidics**
Bilkent University · Ankara, Turkey
• Designed and fabricated microfluidic lasers and waveguides using colloidal semiconductor quantum wells.

Sep 2017–Aug 2018 **Undergraduate Research Assistant, Chemical Engineering**
Middle East Technical University · Ankara, Turkey
• Fabricated cellulose membranes for organic solvent nanofiltration and industrial wastewater applications.

TEACHING EXPERIENCE

Starting Jan 2027 **Assistant Lecturer — Faculty of Engineering**
University of Alberta
Upcoming appointment beginning January 2027.

Jan–Apr 2025 **Course Instructor — CIV E 429, Environmental Engineering Design**
University of Alberta
Senior undergraduate course in drinking water and wastewater treatment facility design for indirect potable reuse; included field trips, design projects, technical reports, and presentations.

Fall 2024 **Biological Processes (ENV E 324)**
University of Alberta
Graduate teaching assistant.

Fall 2024 **Environmental Chemistry (ENV E 220)**
University of Alberta
Graduate teaching assistant.

Fall 2024 **Environmental Water Quality Management (CIV E 525)**
University of Alberta
Graduate teaching assistant.

Winter 2023 and 2024 **Principles of Environmental Modelling and Risk (CIV E 321)**
University of Alberta
Graduate teaching assistant.

Winter 2023 and 2024 **Chemical and Physical Processes (ENV E 325)**
University of Alberta
Graduate teaching assistant.

Fall 2023 **Fluid Mechanics (CIV E 330)**
University of Alberta
Graduate teaching assistant.

Fall 2023 **Municipal Systems (CIV E 421)**
University of Alberta
Graduate teaching assistant.

TEACHING DEVELOPMENT

Apr 2026 **Graduate Teaching and Learning Program — Level 3**
University of Alberta
Pedagogy and Course Design.

Jan 2026 **Graduate Teaching and Learning Program — Level 1**
University of Alberta
Foundations.

SERVICE

2026–Present **Peer Reviewer**
Processes, Applied Sciences, Water, Environment

2026–Present **Publishing Team Lead**
2027 IWA Young Water Professionals Canada Conference

Aug 2026 **Oral Presentation and Poster Judge**
Faculty of Engineering Graduate Research Symposium

Jul 2026 **Volunteer**
5th International Conference on Disinfection and Disinfection Byproducts

TECHNICAL SKILLS

Modelling: Aspen Plus, Aspen HYSYS, BioWin, SolidWorks, WaterGEMS
Data analysis: MATLAB, R, Mathcad, Origin
Analytical methods: HPLC, UV–Vis, TOC, COD, ion chromatography, water-quality analyses
Characterization: XRD, XPS, FTIR, SEM, TEM, EIS, EPR, diffuse reflectance
Languages: English, Arabic, Turkish

AWARDS AND RECOGNITION

Jul 2026 **Best Oral Presentation Award, 5th International Conference on Disinfection and Disinfection Byproducts**

Jun 2026 **Graduate Student Association Academic Travel Grant**
$500

May 2026 **Future Energy Systems: Graduate Student and Postdoctoral Scholar Opportunity Award**
$2,500

May 2026 **Postdoctoral Fellows Association Travel Award**
$500

Nov 2025 **Highly Qualified Professional, Canada First Research Excellence Fund**
$60,000

Sep 2024 **Alberta Innovates Graduate Student Scholarship**
$30,000

Sep 2024 **Alberta Graduate Excellence Scholarship**
$12,000

Sep 2024 **Gordon R Finch Memorial Graduate Scholarship in Environmental Engineering**
$2,700

Sep 2024 **Jacob H Masliyah Graduate Award in Oil Sands Engineering**
$2,000

Sep 2021 **University of Alberta Graduate Recruitment Scholarship**

Sep 2018 **Bilkent UNAM Graduate Scholarship**

Jun 2018 **Middle East Technical University: First in rank certificate**

PROFESSIONAL AFFILIATIONS

May 2026 **Engineer-in-Training (E.I.T.)**
Association of Professional Engineers and Geoscientists of Alberta

May 2025 **Young Water Professional**
International Water Association Canada

PUBLICATIONS AND SCHOLARLY WORK

Peer-Reviewed Journal Articles

1. Kahkeci, J., Sánchez-Montes, I., and Gamal El-Din, M. (2026). Tuning Pyrolysis Temperature of Wheat Straw Biochar Supports for Enhanced Bi2WO6 Photocatalytic Degradation of Stormwater Contaminants in Real Water Matrices. *Journal of Hazardous Materials*, 512, 142241. DOI: 10.1016/j.jhazmat.2026.142241

2. Kahkeci, J., Aoudi, B., Sánchez-Montes, I., and Gamal El-Din, M. (2025). Softwood Biochar-Supported Bi2WO6 for Photocatalytic Degradation of Organic Contaminants in River Water: Role of Pyrolysis Temperature and Surface Functionality. *Journal of Colloid and Interface Science*, 139132. DOI: 10.1016/j.jcis.2025.139132

3. Aoudi, B., Kahkeci, J., Sánchez-Montes, I., Boluk, Y., and Gamal El-Din, M. (2025). Efficient solar-driven degradation of a tire wear pollutant using floating K-doped g-C3N4 photocatalyst in secondary municipal wastewater. *Separation and Purification Technology*, 364, 132523. DOI: 10.1016/j.seppur.2025.132523

4. Aoudi, B., Kahkeci, J., Boluk, Y., and Gamal El-Din, M. (2024). Enhanced g-C3N4 for sustainable solar degradation of 1,3-diphenylguanidine (DPG) in wastewater: Investigating the effects of precursor, temperature, and potassium doping. *Chemical Engineering Journal*, 487, 150665. DOI: 10.1016/j.cej.2024.150665

5. Kahkeci, J., Aoudi, B., Sánchez-Montes, I., and Gamal El-Din, M. (2024). Engineered biochar supported bismuth tungstate: Unveiling the influence of precursor concentrations and biochar dosage for the solar photocatalysis of 1,3-diphenylguanidine in secondary municipal effluent. *Chemical Engineering Journal*, 483, 149142. DOI: 10.1016/j.cej.2024.149142

6. Kahkeci, J. and Gamal El-Din, M. (2023). Biochar-supported photocatalysts: Performance optimization and applications in emerging contaminant removal from wastewater. *Chemical Engineering Journal*, 476, 146530. DOI: 10.1016/j.cej.2023.146530

7. Maskoun, J., Gheshlaghi, N., Isik, F., Delikanli, S., Erdem, O., Erdem, E. Y., and Demir, H. V. (2021). Optical microfluidic waveguides and solution lasers of colloidal semiconductor quantum wells. *Advanced Materials*, 33(10), 2007131. DOI: 10.1002/adma.202007131

Manuscripts Submitted

1. Aoudi, B., Kahkeci, J., and Gamal El-Din, M. (2026). Photocatalytic treatment of oil sands process water: recent advances, reactive pathways, treatment performance, and future opportunities. *Current Opinion in Chemical Engineering*.

2. Mokarizadeh, H., Aoudi, B., Kahkeci, J., Ganiyu, S., and Gamal El-Din, M. (2026). Solar-Driven Photocatalysis for Water and Wastewater Treatment: Material Trends, Mechanistic Insights, and Emerging Challenges. *Environmental Science and Pollution Research*.

Conference Presentations

1. Kahkeci, J., Aoudi, B., Sánchez-Montes, I., and Gamal El-Din, M. (Jul 2026). Tuning softwood biochar pyrolysis temperature to enhance solar-driven Bi2WO6 photocatalysis for organic micropollutant removal in river water. 5th International Water Association (IWA) International Conference on Disinfection and Disinfection By-Products · Shanghai, China.

2. Gamal El-Din, M., Usman, M., Feng, Q., Bekele, A., Kahkeci, J., Li, Y., Fan, Y., Arslan, M., Jaffe, B., Jessup, R. W., and Bremer, E. (Jul 2026). Biochar and Torrefied Biomass Soil Amendments: Influence on Nutrient Dynamics in Clayey and Sandy Soils. 1st International Conference on Land Degradation and Restoration · Valencia, Spain.

3. Aoudi, B., Kahkeci, J., Boluk, Y., and Gamal El-Din, M. (Aug 2024). Advanced techniques for wastewater treatment: Enhancing photocatalytic efficiency of graphitic carbon nitride. IWA World Water Congress and Exhibition 2024 · Canada.

4. Aoudi, B., Kahkeci, J., Boluk, Y., and Gamal El-Din, M. (Jun 2024). Sustainable water treatment: Photocatalytic degradation of 1,3-diphenylguanidine (DPG) using graphitic carbon nitride. Canadian Society for Civil Engineering Annual Conference · Niagara Falls, ON.

5. Kahkeci, J., Aoudi, B., Sánchez-Montes, I., and Gamal El-Din, M. (Jun 2024). Enhancing Solar Photocatalysis of Emerging Contaminants in Wastewater using Biochar Supported Bismuth Tungstate: Effects of Biochar Supports and Synthesis Parameters. Canadian Society for Civil Engineering Annual Conference · Niagara Falls, ON.

6. Kahkeci, J. and Sarathy, S. M. (Jun 2021). Green Hydrogen from Saline Water. King Abdullah University of Science and Technology Research Conference: Near-Zero Carbon Combustion Technology · Thuwal, Saudi Arabia.

7. Kahkeci, J. and Culfaz-Emecen, P. Z. (May 2018). Fabrication of Cellulose Membranes for Organic Solvent Nanofiltration via Alkaline Hydrolysis of Cellulose Acetate. Gazi University Chemical Engineering Conference · Ankara, Turkey.